

Report: Backup & Recovery

1. Introduction

The integrity and availability of Active Directory are critical for any domain-based enterprise infrastructure. To ensure resilience against unforeseen data loss or configuration errors, administrators must be equipped with strong backup and recovery strategies. This task was focused on implementing a system state backup on a domain controller and validating the recovery process through simulation. It included enabling critical Active Directory features like the Recycle Bin, simulating an Organizational Unit (OU) deletion, and then restoring the OU using recycle bin and another OU by authoritative restoration procedure. These steps helped verify the preparedness of the Active Directory environment to handle accidental deletions and maintain continuity of operations in case of such incidents. The process was carried out in a virtual lab environment with AD-DS configured on a Windows Server 2022 machine.

2. Objective

The primary objective of this task was to ensure that backup and recovery mechanisms for Active Directory are properly configured and tested. It aimed to develop a working understanding of how system state backup captures key AD components, how the AD Recycle Bin can be utilized to recover deleted objects, and how authoritative restore can be used in scenarios where the Recycle Bin is not sufficient. The task also focused on building the ability to carry out these operations through command-line tools in DSRM, reinforcing the ability to recover from critical failures or misconfigurations. Another key objective was to confirm that all recovery steps work without causing inconsistency in the domain database and that restored objects retain their original integrity and properties.

3. Tools used

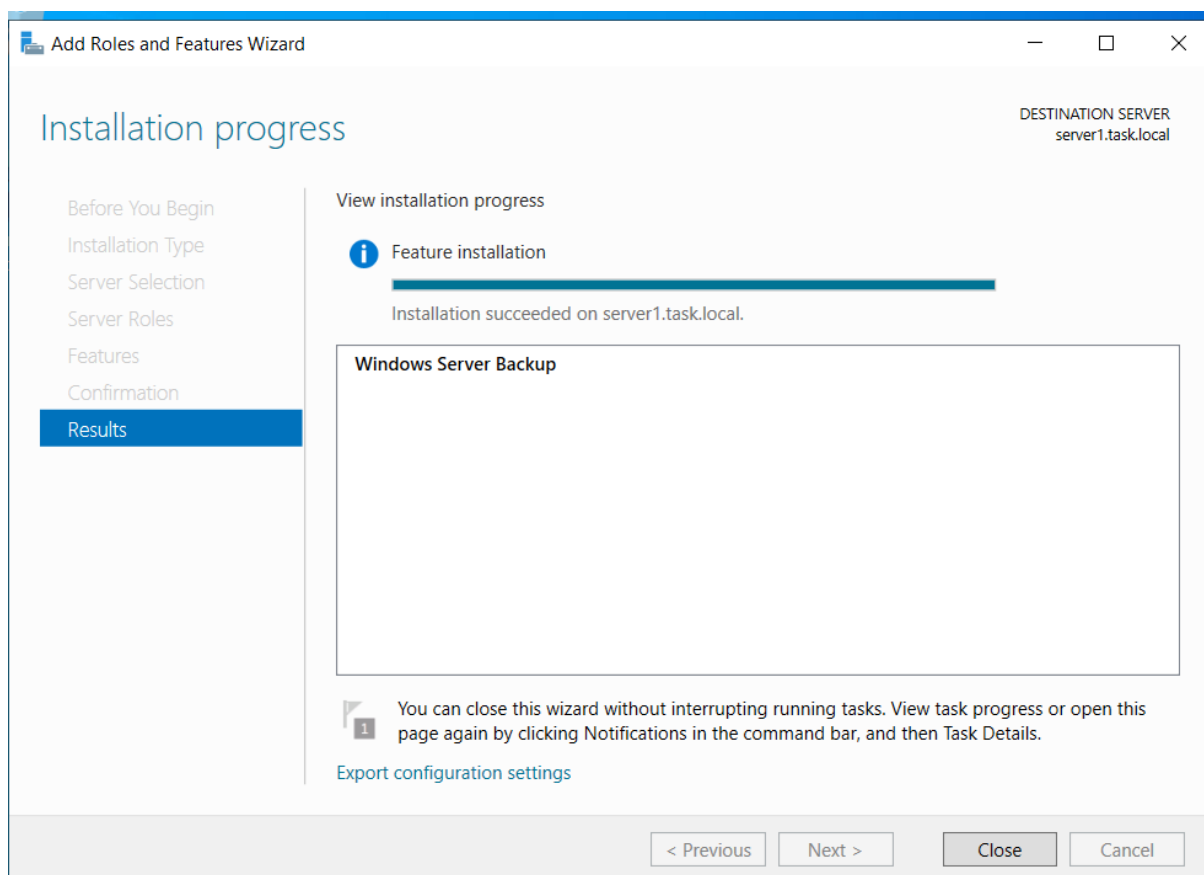
- Windows Server Backup (wbadmin)
- PowerShell

- ntdsutil
- Command Prompt (Administrator)

4. Methodology

4.1 System State Backup

The system state backup was initiated from the Windows Server. A backup target volume was configured and verified. The command `wbadmin start systemstatebackup -backupTarget:E: -quiet` was executed in PowerShell. This command started the system state backup and ensured the full configuration of the domain controller was captured. The backup process was confirmed through completion logs and successful system state operation messages.



(Installing windows server backup)

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\Administrator> wbadmin start systemstatebackup -backupTarget:E: -quiet
PS C:\Users\Administrator> wbadmin start systemstatebackup -backupTarget:E: -quiet
wbadmin 1.0 - Backup command-line tool
(C) Copyright Microsoft Corporation. All rights reserved.

Starting to back up the system state [5/13/2025 6:02 PM]...
Retrieving volume information...
This will back up the system state from volume(s) (EFI System Partition),(C:),(\?\Volume{c3c56d78-5dc4-4a5a-ae4f-91eea9b72198}\) to E:..
Creating a shadow copy of the volumes specified for backup...
Please wait while system state files to back up are identified.
This might take several minutes...
Please wait while system state files to back up are identified.
This might take several minutes...
Please wait while system state files to back up are identified.
This might take several minutes...
Found (261) files.
Found (268) files.
Found (2409) files.
Found (4348) files.
Found (11373) files.
Found (14270) files.
Found (22899) files.
```

(Initialize backup)

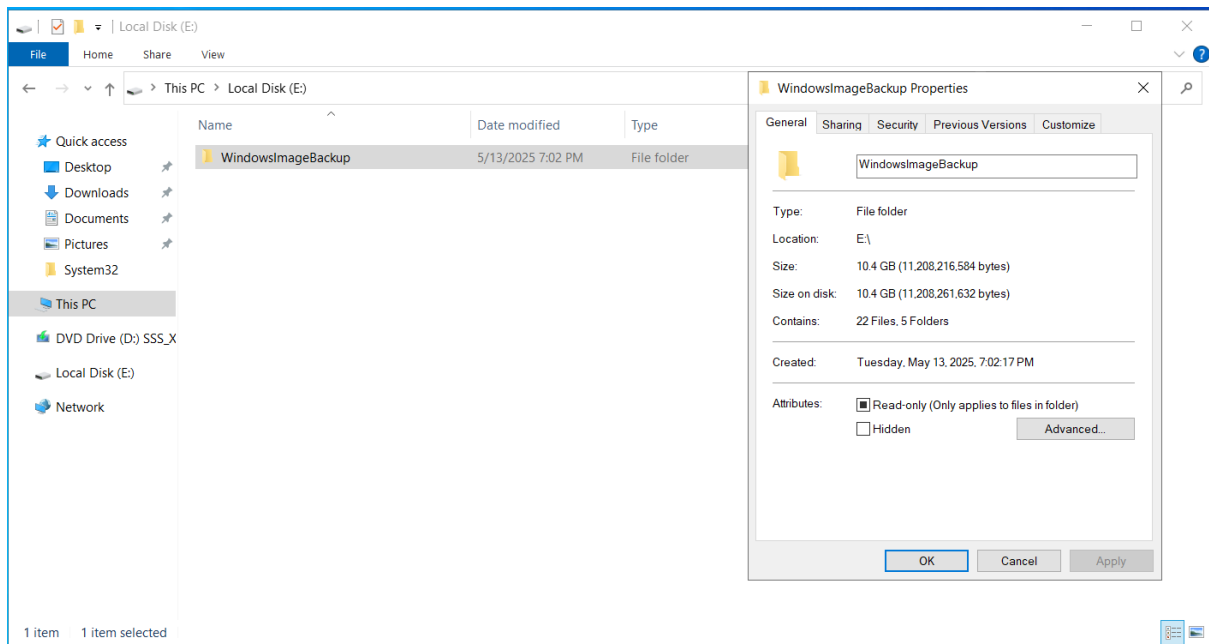
```
Select Administrator: Windows PowerShell

Overall progress: 37%.
Currently backing up files reported by 'System Writer'...
Overall progress: 38%.
Currently backing up files reported by 'System Writer'...
Overall progress: 39%.
Currently backing up files reported by 'System Writer'...
Overall progress: 41%.
Currently backing up files reported by 'System Writer'...
Overall progress: 43%.
Currently backing up files reported by 'System Writer'...
Overall progress: 44%.
Currently backing up files reported by 'System Writer'...
Overall progress: 45%.
Currently backing up files reported by 'System Writer'...
Overall progress: 46%.
Currently backing up files reported by 'System Writer'...
Summary of the backup operation:
-----

The backup operation successfully completed.
The backup of the system state successfully completed [5/13/2025 2:18 PM].
Log of files successfully backed up:
C:\Windows\Logs\WindowsServerBackup\Backup-13-05-2025_13-32-10.log

PS C:\Users\Administrator>
PS C:\Users\Administrator>
```

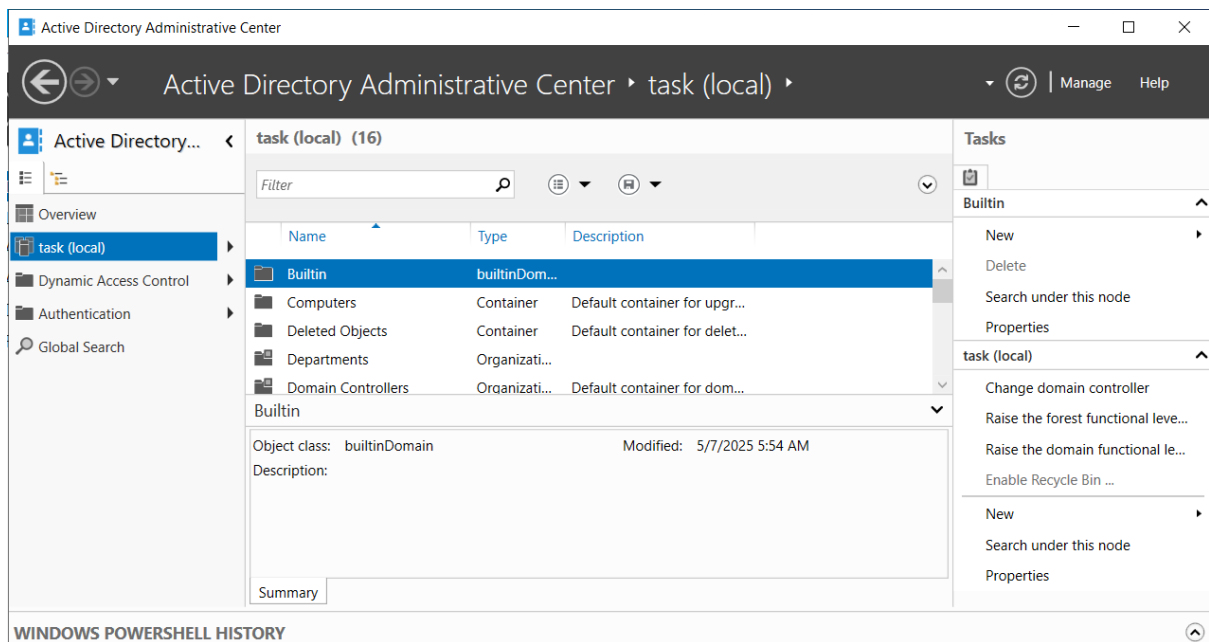
(Backup completed)



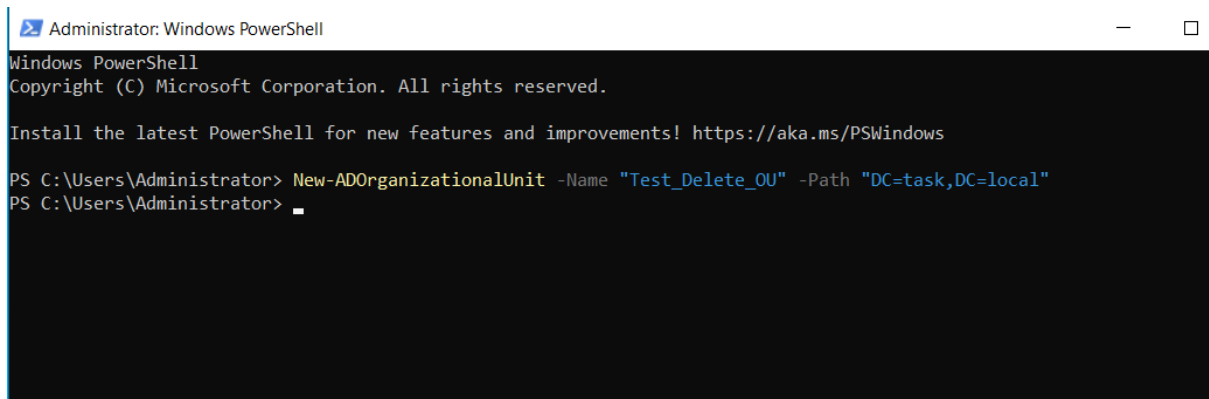
(Backup stored in disk E which we created for this task)

4.2 Simulate Deletion of OU

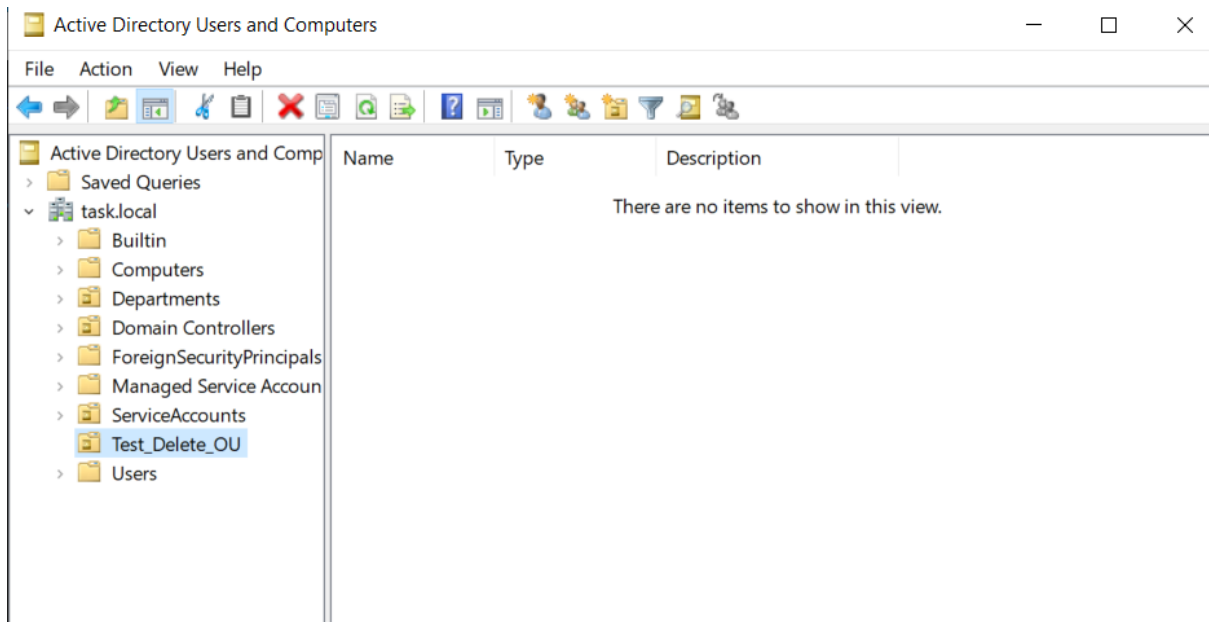
Before deletion, the Active Directory Recycle Bin was enabled using the PowerShell command `Enable-ADOptionalFeature -Identity 'Recycle Bin Feature' -Scope ForestOrConfigurationSet -Target "task.local"`. An Organizational Unit named `Test_Delete_OU` was created and its accidental deletion protection setting was removed manually through the GUI. Once confirmed, the OU was deleted using the `Remove-ADOrganizationalUnit` PowerShell command with the `-Recursive` parameter.



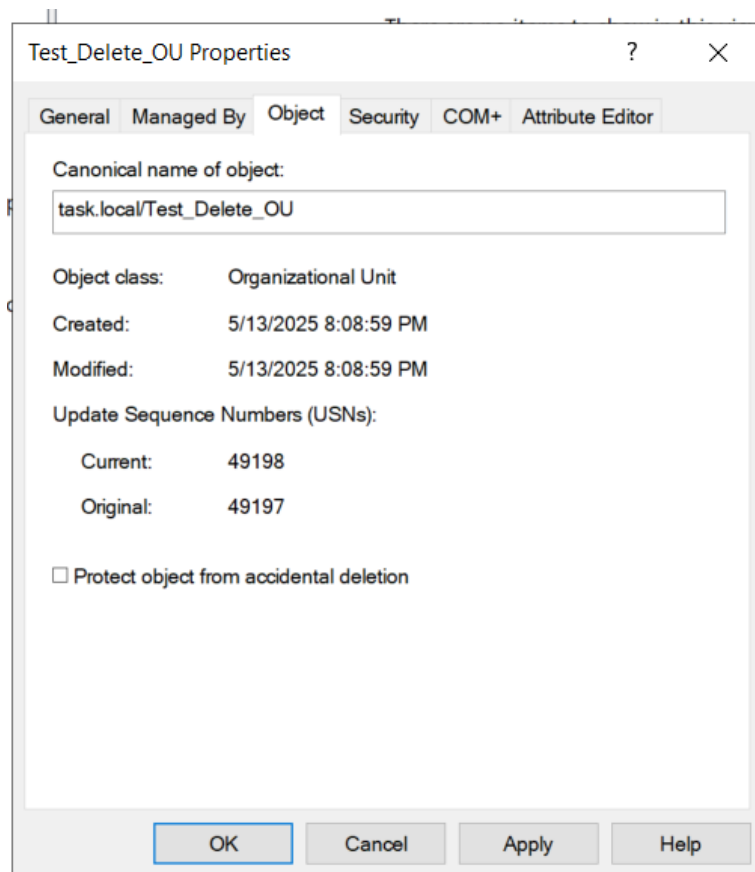
(Recycle bin has been enabled)



(Created a new OU)



(Successfully created)

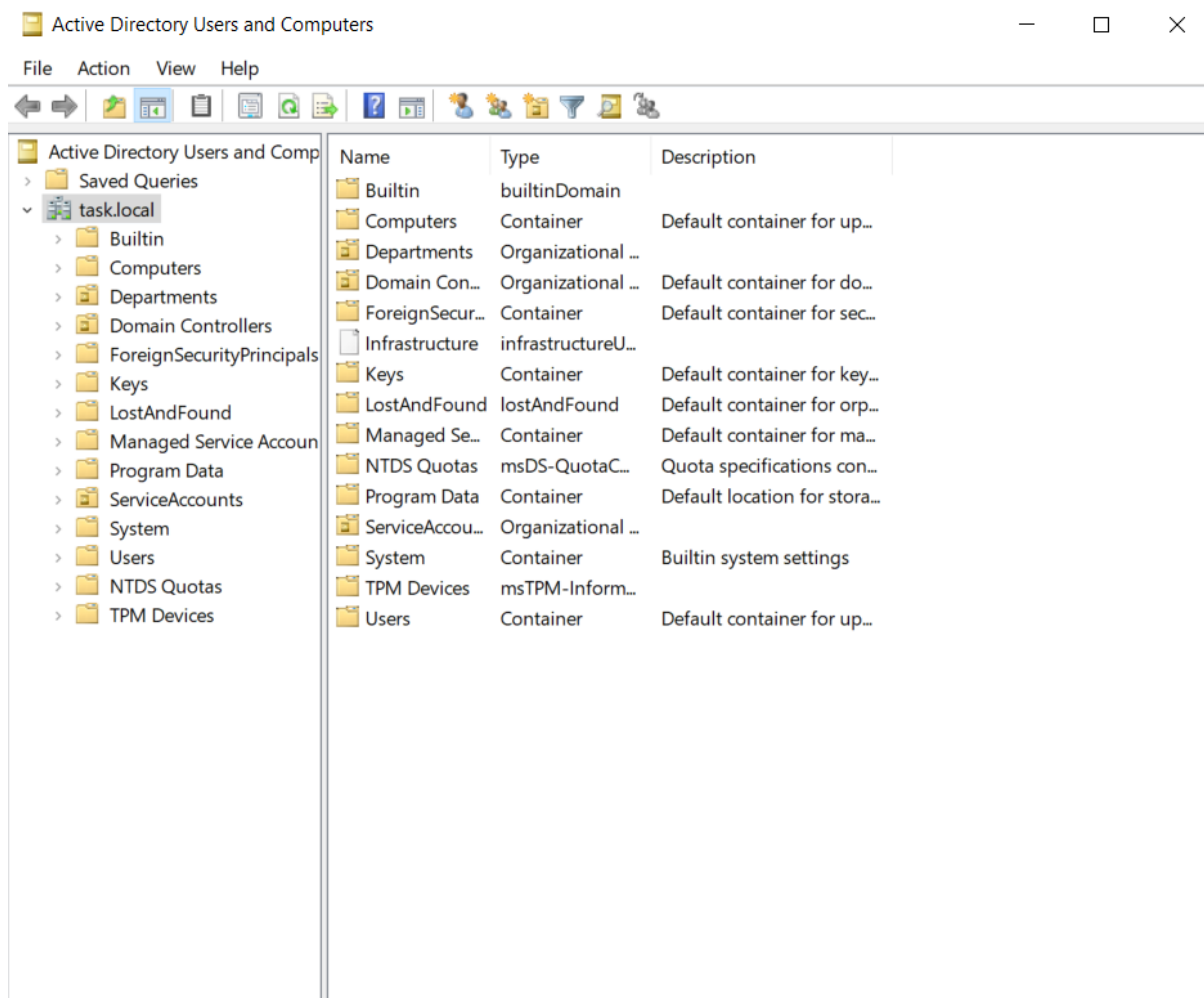


(Disabled the accidental deletion)

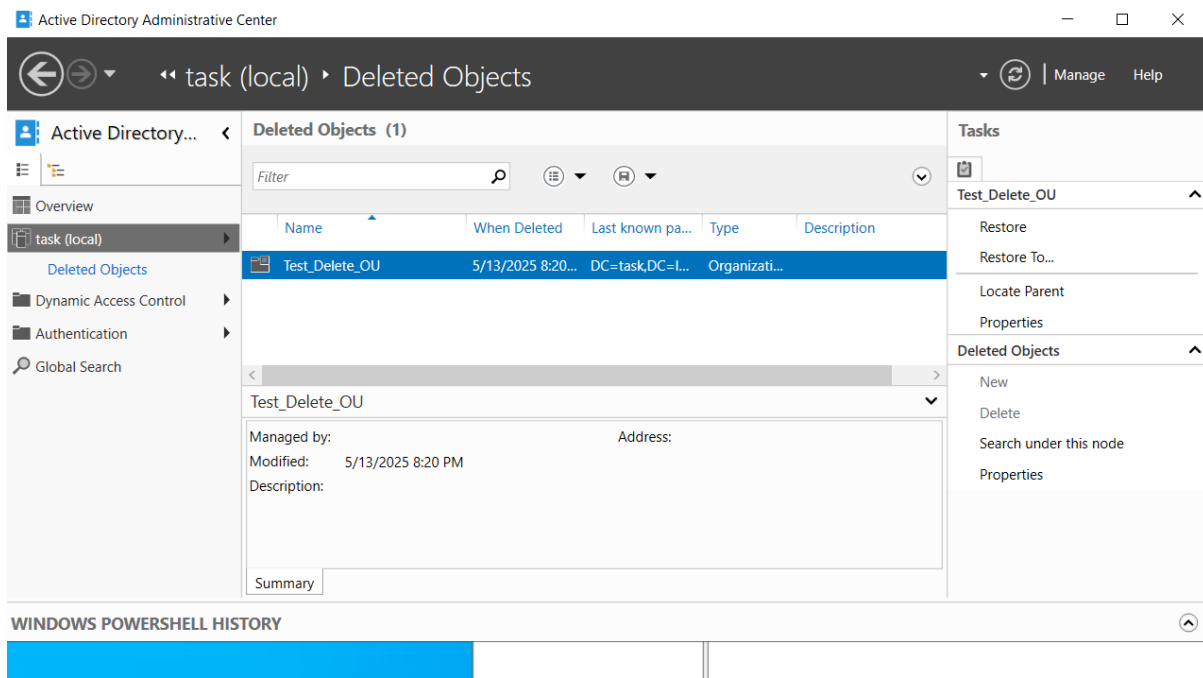
```
Administrator: Windows PowerShell
PS C:\Users\Administrator> Remove-ADOrganizationalUnit -Identity "OU=Test_Delete_OU,DC=task,DC=local" -Recursive

Are you sure you want to remove the item and all its children?
Performing recursive remove on Target: 'OU=Test_Delete_OU,DC=task,DC=local'.
[Y] Yes [A] Yes to All [N] No [L] No to All [S] Suspend [?] Help (default is "Y"): y
PS C:\Users\Administrator>
```

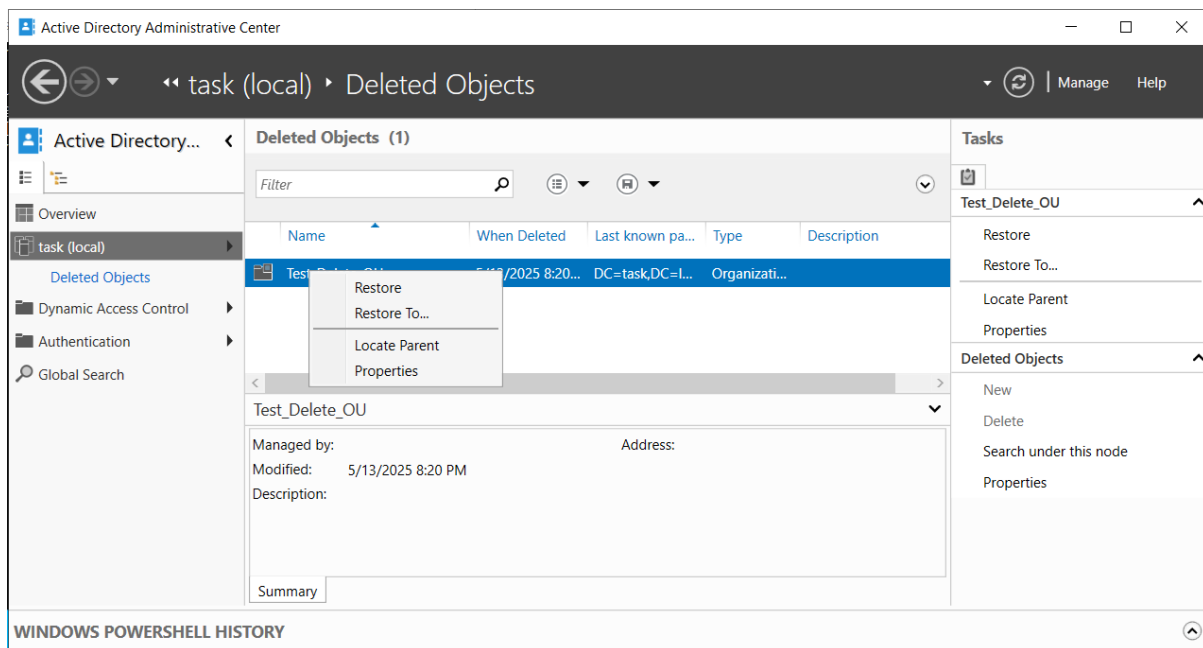
(Command to delete executed)



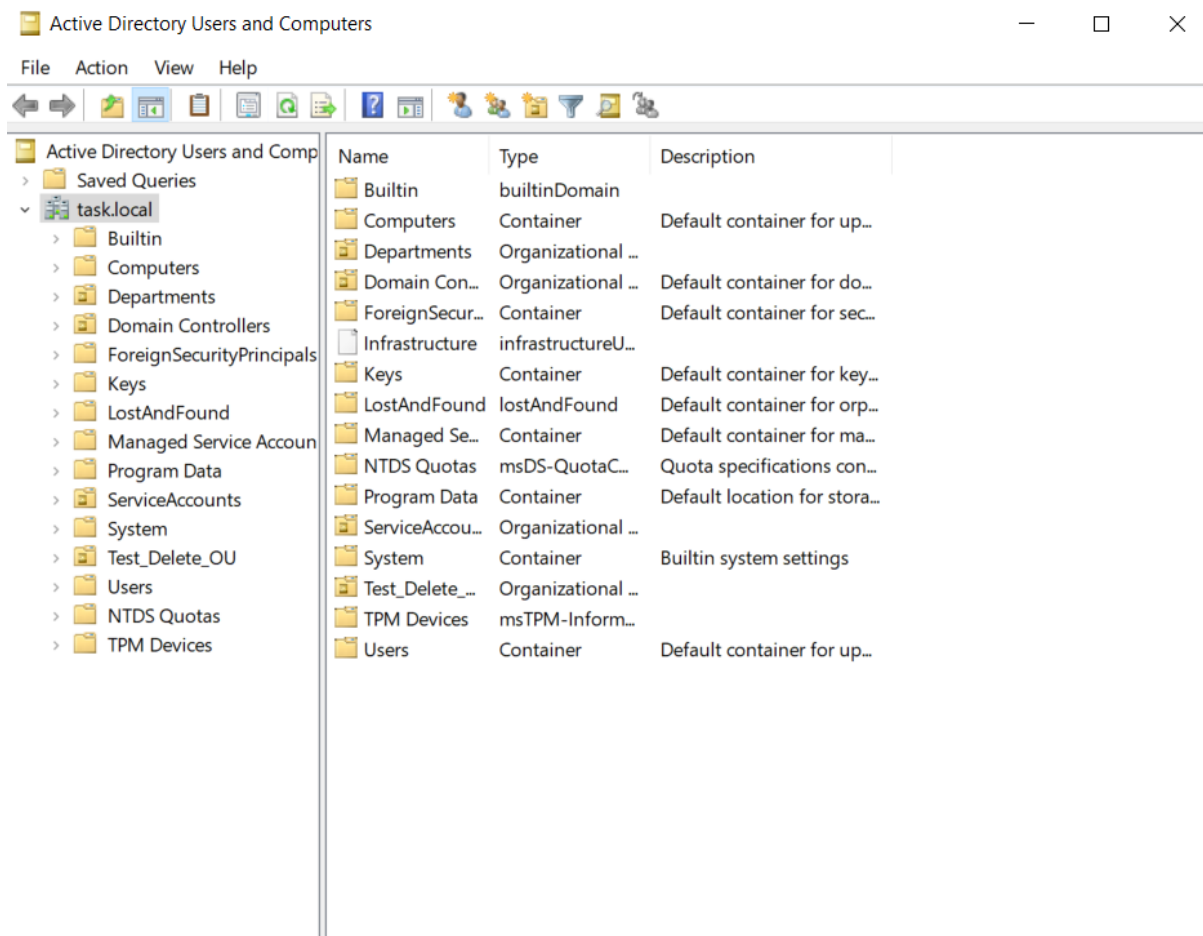
(Deleted successfully)



(It is showing in our recycle bin)



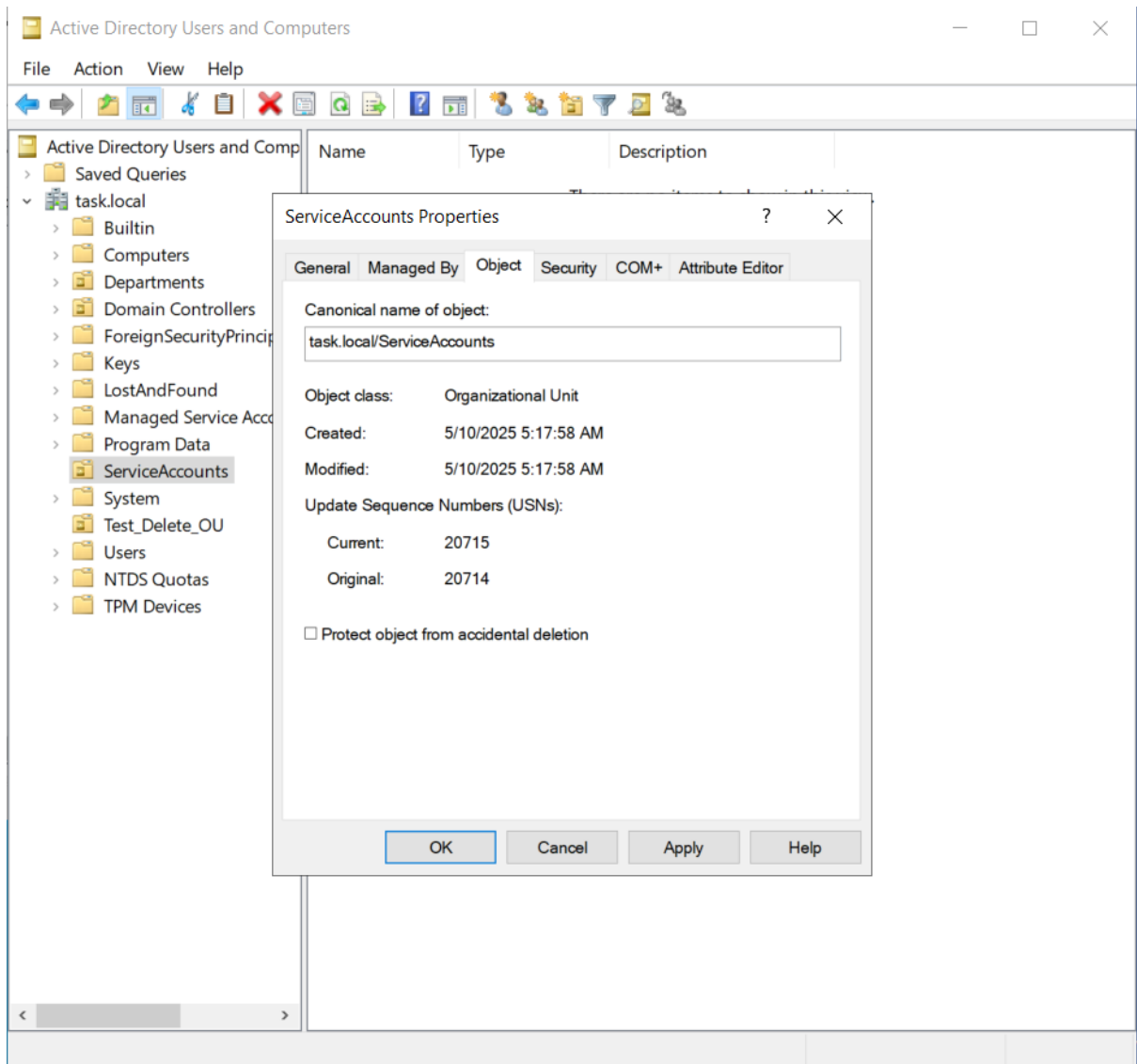
(Let's restore it)



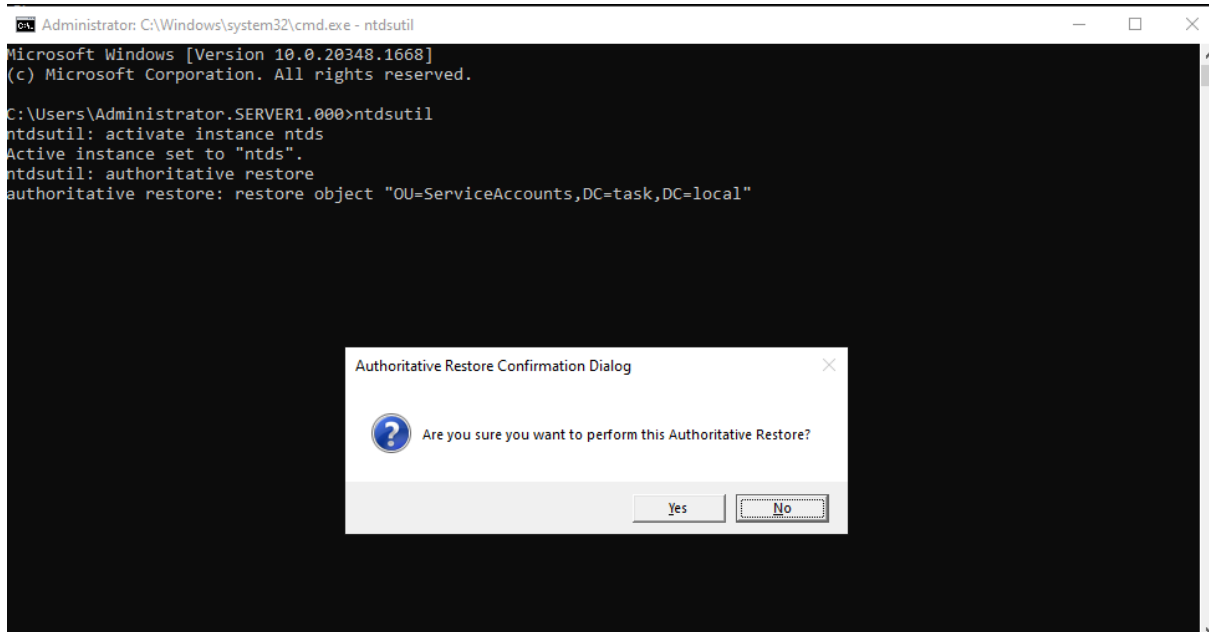
(Restored successfully)

4.3 Authoritative Restore of Deleted OU

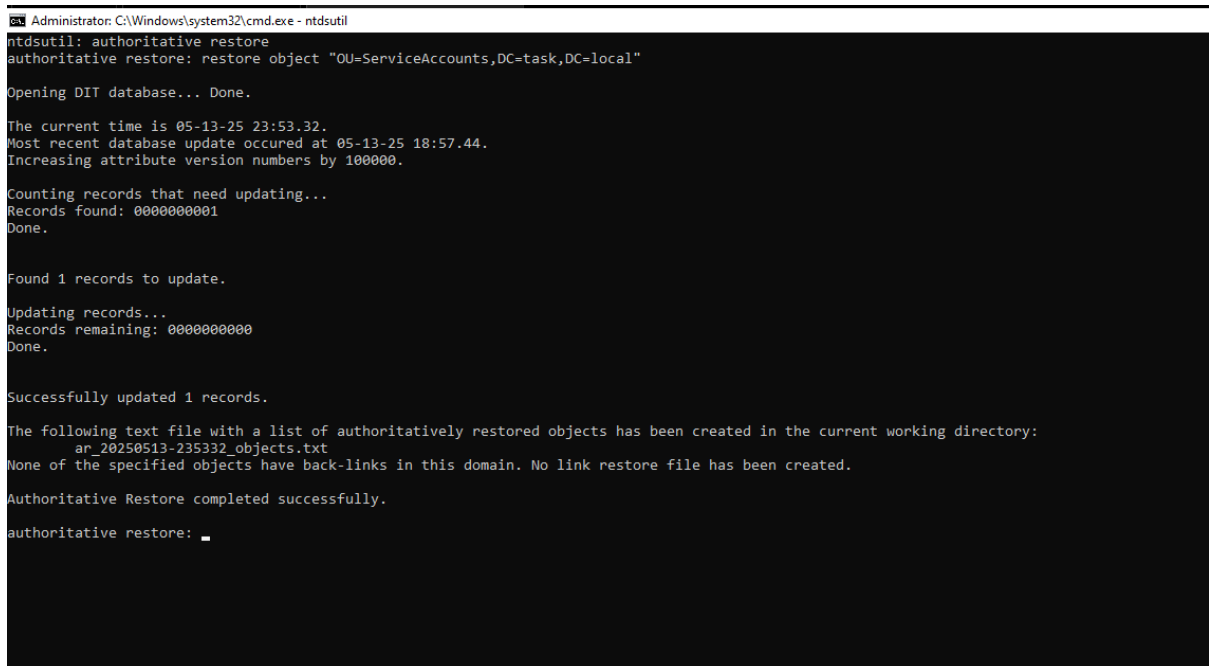
The system was restarted into Directory Services Restore Mode (DSRM) to carry out the authoritative restore. Within the DSRM session, the ntdsutil tool was used to activate the authoritative restore context. The restore command restore object "OU=ServiceAccounts,DC=task,DC=local" was executed, marking the specified OU for authoritative restoration. After restoration, the system was rebooted normally and the restored OU was verified using the Active Directory Users and Computers console.



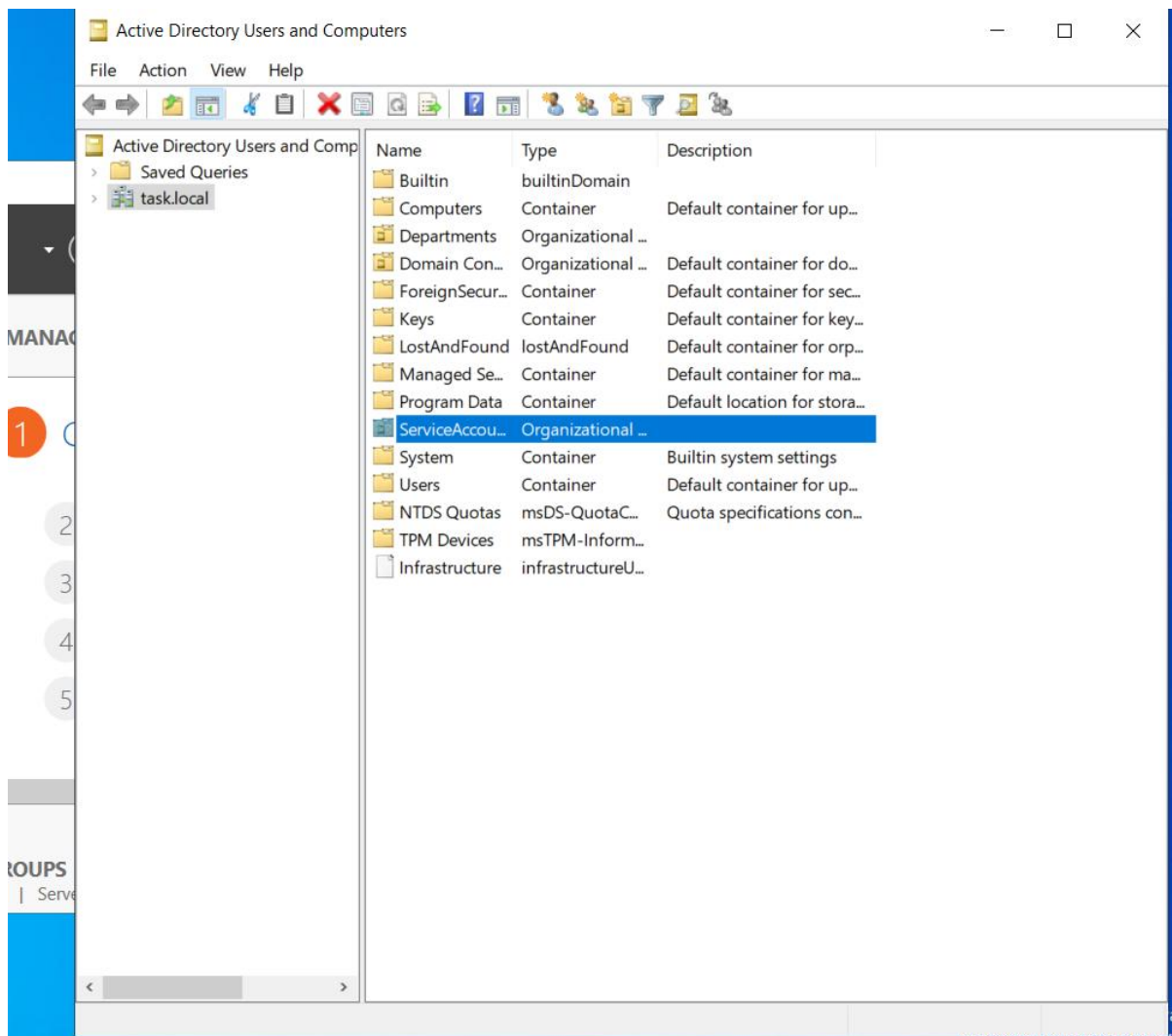
(Let's delete service accounts)



(Authoritative restore initialize)



(Successfully restored)



(We got our deleted ServiceAccounts back)

5. Recommendations

- Ensure sufficient disk space before initiating system state backups.
- Perform regular system state backups and store them on a secure volume.
- Enable AD Recycle Bin early in deployment for object recovery capability.
- Practice and document the DSRM login process for emergency use.
- Use authoritative restore only for objects that existed during the last backup.
- Maintain detailed logs of backup and recovery operations.

6. Conclusion

The tasks successfully validated the process of backing up and restoring Active Directory components using native Windows tools and best practices. System state backup using wbadmin was performed and confirmed, ensuring that the AD environment can be restored in the event of a system failure or misconfiguration. The Recycle Bin feature was enabled to support basic object restoration, and an OU deletion was simulated to test the recovery mechanism. Additionally, an authoritative restore was completed in DSRM mode to recover an OU that was created prior to the backup but deleted afterward. The recovered OU was verified for consistency and completeness. These steps demonstrate the importance of proactive disaster recovery preparation and prove the environment's ability to handle and recover from critical AD object loss scenarios.